Unit: Unit 3: Functions
Topic: Function Transformations

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. The function $f(x) = x^2$ is vertically shifted 2 units down. What is the new function?
(A) $f(x) = x^2 - 2$
(B) $f(x) = x^2 + 2$
(C) $f(x) = x^2 + 1$
(D) $f(x) = (x - 2)^2$
(E) $f(x) = (x + 2)^2$

Q2. A company's stock price is modeled by $f(x)$, where x is the number of days. If $f(x)$ is reflected over the y-axis, what does this represent?
(A) The company's stock price increasing
(B) The company's stock price decreasing
(C) The company's stock price over time in reverse order
(D) The company's stock price over a different time period
(E) The company's stock price over the same time period with no change

Q3. The function $f(x) = x^2$ is horizontally shifted 3 units to the left. What is the new function?
(A) $f(x) = (x + 3)^2$
(B) $f(x) = (x - 3)^2$
(C) $f(x) = (x - 2)^2$
(D) $f(x) = x^2 + 2$
(E) $f(x) = x^2 - 2$

Q4. A savings account earns interest according to the function $f(x) = 2x$, where x is the amount of money deposited. If the interest rate is doubled, what is the new function?
(A) $f(x) = x^2$
(B) $f(x) = 4x$
(C) $f(x) = x + 2$
(D) $f(x) = 2x + 1$
(E) $f(x) = x - 2$

Q5. The function $f(x) = x^2$ is stretched vertically by a factor of 2. What is the new function?
(A) $f(x) = 2x^2$
(B) $f(x) = (2x)^2$
(C) $f(x) = x^2 + 2$
(D) $f(x) = x^2 - 2$
(E) $f(x) = x^2/2$

Q6. A tax rate is modeled by the function $f(x)$, where x is the amount of income. If $f(x)$ is compressed horizontally by a factor of 2, what does this represent?
(A) The tax rate increasing
(B) The tax rate decreasing
(C) The tax rate applying to twice the amount of income
(D) The tax rate applying to half the amount of income
(E) The tax rate remaining the same

Q7. The function $f(x) = x^2$ is reflected over the x-axis. What is the new function?
(A) $f(x) = -x^2$
(B) $f(x) = x^2$
(C) $f(x) = -x$
(D) $f(x) = x$
(E) $f(x) = x^3$

Q8. A company's profit is modeled by the function $f(x)$, where x is the number of units sold. If $f(x)$ is shifted 2 units up, what does this represent?
(A) The company's profit increasing by 2 for each unit sold
(B) The company's profit decreasing by 2 for each unit sold
(C) The company's profit remaining the same
(D) The company's profit increasing by 2 regardless of the number of units sold
(E) The company's profit decreasing by 2 regardless of the number of units sold

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Open Ended Questions (FRQ)

Q9. Describe how the function $f(x) = x^2$ would change if it were stretched vertically by a factor of 3 and then shifted 1 unit to the right. Show the new function and provide a brief explanation.

Q10. Suppose a stock's price is modeled by the function $f(x) = 2x$. If the function is reflected over the y-axis and then compressed horizontally by a factor of 2, describe the new function and its implications for the stock's price. Show the new function and provide a brief explanation.

Chef's Tips

- Emphasize the importance of understanding function transformations and their real-world applications.
- Use visual aids to help students visualize the transformations.
- Encourage students to check their work by graphing the original and transformed functions.